Day 6 – Foundation Models & Multimodal AI

1. Introduction

Modern Generative AI systems like **ChatGPT, Gemini, Claude, Llama, and Midjourney** are built on a new class of models called **Foundation Models**.

These models are:

- Large
- Pretrained on massive data
- General-purpose
- Adaptable to many tasks

GPT-5
BERT → Ex:
OPUS 4.6

Later, AI evolved further into **Multimodal Models**, which can understand **text, images, audio, video, and more simultaneously**.

Example:

Training →
Website
Books
Research papers
Code
image & video

You show AI a **photo of food**, and it can:

- describe it
- generate recipe
- suggest calories
- generate cooking video.

Ex :- Bert → 2018 → train → 340 million parameters & 16GB dataset

2023 → OpenAI → GPT-4 → 170 billion parameters 45 GB dataset

2. What is a Foundation Model?

Definition

A **Foundation Model** is a **large pretrained AI model trained on massive unlabeled datasets that can be adapted to many downstream tasks.**

These models act as a **base or foundation for multiple AI applications.**

Simple Definition

Foundation Model = **Large Pretrained Model + Fine-tuning for many tasks**

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3. Why Foundation Models Were Introduced

Before foundation models, AI worked like this:

Old AI approach:

Task-specific models

Example

Task	Model
Spam Detection	Model A
Translation	Model B
Sentiment	Model C

Problem:

- Need separate models
- Large engineering cost
- No knowledge sharing

Solution

Foundation models.

Train **one massive model** and adapt it to many tasks.

4. Foundation Model Concept (Core Idea)

The idea is:

```
Large Internet Data
        ↓
 Pretraining Phase
        ↓
 Large Foundation Model
        ↓
 Fine-tuning / Prompting
        ↓
 Many Applications
```

Applications include:

- Chatbots
- Coding assistants
- Image generation
- Video generation
- Robotics
- Search engines
- Medical diagnosis

5. Foundation Model Architecture

Most foundation models today are based on **Transformer architecture**.

Architecture Diagram

```
            Massive Data
        (text + images + code)

                ↓
           Tokenization

                ↓
        Transformer Architecture
        (Self Attention Layers)

                ↓
           Representation Learning

                ↓
         Pretrained Foundation Model
```

Characteristics of Foundation Models

1. Large Scale

Parameters:
- GPT models: trillions of parameters

Large data:
- trillions of tokens.

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2. Self-supervised learning

Training without manual labels.

Example task:

Predict masked words.

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3. Transfer Learning

Knowledge learned from one task helps another.

Example:

A model trained on internet text can:
- translate
- summarize
- answer questions.

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4. Emergent Abilities

New capa'bilities appear automatically.

Example:
- reasoning
- coding
- math solving

What is Multimodal AI?

Now comes the next revolution.

Traditional LLMs process **only text**.

But humans perceive the world through multiple senses:

- Vision
- Sound
- Language
- Motion

Multimodal AI mimics this ability.

Definition

A **multimodal AI model** can process and generate **multiple types of data simultaneously**, including:

- Text
- Images
- Audio
- Video
- Sensor data

These systems integrate information across modalities to produce out

12. Modalities in AI

Modalities = types of data.

Modality	Example
Text	ChatGPT
Image	DALL-E
Audio	Whisper
Video	Sora
Sensor	robotics

Multimodal models combine them.

5. Example of Multimodal AI

Example 1 – Image + Question

Input:

- A picture of a dog
- Question: *What animal is this?*

Output:

- "This is a dog."

Here the model understands:

- the **image**
- the **text question**

This is a multimodal task.

Example 2 – Image Description

Input:

- A photo of a beach

Output:

```
code

A beautiful beach with blue water and white sand.
```

The AI converts **image → text**.